

Summary Output:

Genome ID	Contigs	Size(bp)	GC%	CodDen	CDS	Known	Hypo	Hypo%	Pseudo	tRNA	rRNA	CRISPR	sORF
GCF_000007545.1_ASM754v1_genomic	1	4791961	52.1	88.4	4618	4132	486	10.5%	0	80	22	1	7
GCF_000020745.1_ASM2074v1_genomic	3	4823887	52.2	88.7	4533	4270	263	5.8%	0	86	22	2	8
GCF_000020885.1_ASM2088v1_genomic	2	4836638	52.0	88.7	4496	4226	270	6.0%	0	86	22	2	8
GCF_000020925.1_ASM2092v1_genomic	2	4917459	52.1	88.6	4669	4326	343	7.3%	0	86	22	0	9
GCF_000022165.1_ASM2216v1_genomic	2	4964097	52.2	88.8	4639	4383	256	5.5%	0	88	22	2	9
GCF_000170215.1_ASM17021v1_genomic	2	4726474	52.3	88.9	4336	4143	193	4.5%	0	80	22	2	10
GCF_000171415.1_ASM17141v1_genomic	63	4948011	52.1	88.4	4678	4369	309	6.6%	0	65	2	5	9
GCF_000171515.1_ASM17151v1_genomic	50	4793325	52.3	88.6	4444	4189	255	5.7%	0	85	33	2	9
GCF_000171535.2_ASM17153v2_genomic	5	4876885	52.1	88.7	4579	4310	269	5.9%	0	86	22	2	9

Summary Statistics

Genome size range : 4726474.0 - 4964097.0 bp
Mean genome size : 4853193.0 bp
GC content range : 52.0 - 52.3 %
CDS range : 4336.0 - 4678.0 genes
Mean CDS / genome : 4554.7
Hypothetical range : 4.5 - 10.5 %
Mean hypothetical% : 6.4 %

Genome Annotation Reflection:

Genome annotation of nine *Salmonella enterica* serovars using Bakta produced a suite of metrics that collectively characterize the functional content and completeness of each assembly. The metrics reported included: total CDS count, hypothetical protein percentage, tRNA and rRNA counts, CRISPR array presence, and genome size. These metrics were selected as Bakta will run all these by default, and testing all metrics will be helpful for comparing genomes. Genome size and GC content serve as foundational quality indicators. Across the nine assemblies, genome sizes ranged from 4.73 to 4.96 Mb with a GC content between 52.0 and 52.3%, both consistent with expected *S. enterica* genome size ranges.

The proportion of hypothetical proteins is one of the most significant metric reported. The mean hypothetical percentage across the nine serovars was 6.4%, but individual values ranged from 4.5% (GCF_000170215) to 10.5% (GCF_000007545). Because these assemblies were originally published in the early 2000s and re-annotated using Bakta against a current database, the hypothetical fractions reported likely underestimate those that would have been observed when Fricke et al. published their 2011 comparative analyses; genes that were uncharacterized then may now have database entries.

tRNA and rRNA counts serve as indirect markers of assembly completeness. Most assemblies in this dataset recovered 22 rRNA annotations, consistent with the expected operon count for *S. enterica*. One notable exception is GCF_000171515, which returned 33 rRNA annotations, which is likely inflated due to assemblies being constructed using older sequencing technology and repetitive regions likely being poorly resolved without long-reads to help polish the assembly. Similarly, GCF_000171415, the most fragmented assembly with 63 contigs, may have gene counts affected by genes split across fragmented contigs, a known limitation of fragmented assemblies that can slightly distort CDS totals and functional summaries.

CRISPR array counts, which reported 0 to 5 across serovars, mediate phage resistance in *Salmonella*. Interestingly, GCF_000020925 had a CRISPR array count of zero, in theory this might lead this strain to be more susceptible to phages, but as the assemblies are older, this may be due to sequencing or assembly errors. These metrics will directly inform how downstream analyses are interpreted; a serovar with a high hypothetical fraction should be treated with appropriate caution when drawing conclusions from functional annotation alone, and the fragmented assembly of GCF_000171415 should be flagged as a potential source of noise in any analysis sensitive to gene boundary accuracy.